

GREEN SKILLS USING AI TECHNOLOGIES

GTU – Edunet Foundation Internship Program

Name: Vedant Patel

Enrollment No.: 241263107039

Stream: Computer Engineering

College: SAL Engineering & Technical Institute (SETI),
Ahmedabad

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Introduction

What are Green Skills?

Definition

Green Skills are the knowledge, abilities, and practices that help protect the environment, conserve natural resources, and support sustainable development.

Why are Green Skills Important?

Protect the Environment

- Reduce pollution and greenhouse gas emissions.

Create Green Jobs

- Open career opportunities in renewable energy, waste management, electric vehicles, and environmental conservation.

Role of AI

- AI improves waste management, predicts climate changes, optimizes energy usage, and supports sustainable agriculture.



Sustainable Development Goals (SDGs)

What are the SDGs?

The Sustainable Development Goals (SDGs) are 17 global goals established by the United Nations in 2015 to achieve a better and more sustainable future by 2030.

They focus on solving global challenges such as:

- Climate Change
- Environmental Protection
- Clean Water & Sanitation
- Quality Education
- Reducing Inequality
- Global Partnerships



What I Learned

- Working together creates greater environmental impact.
- AI and technology help organizations share data and solve environmental problems.
- Partnerships encourage innovation and sustainable development.

What I Learned from the Data Centre Documentary

What is a Data Centre?

A Data Centre is a facility that stores, processes, and manages digital data. It powers cloud services, websites, AI applications, online banking, and streaming platforms.

Key Learnings

High Energy Consumption

- Data centres consume large amounts of electricity to run servers 24/7.

Cooling Systems

- Servers generate heat, so advanced cooling systems are required to maintain performance.

Environmental Impact

- Traditional data centres can produce significant carbon emissions if powered by fossil fuels.

Role of AI

- AI helps optimize cooling, monitor energy usage, predict equipment failures, and improve overall efficiency.

Government Website Activity – CO₂ Emission Calculator (e-AMRIT)

Should I shift to electric vehicle?

Vehicle Segment ⓘ	4W (Four-Wheeler) ▼	Comparing Vehicle Fuel Type ⓘ	PETROL ▼
Journey Distance ⓘ	20 km	Journey Frequency ⓘ	Daily ▼
Annual Journey Distance ⓘ	7300 km	CO2 emissions factor ⓘ	179.94 g/km
Comparable EV Battery Capacity ⓘ	40 kWh	EV Range ⓘ	250 km
Battery needed for total journey ⓘ	1168.00 kWh	Electricity CO2 emissions factor ⓘ	850 g/kWh
Lifecycle of the Vehicle ⓘ	15 years		

Calculate

Vehicle Details Used :

Vehicle Type: 4W (Four-Wheeler) , Fuel: Petrol

Daily Distance: 20 km , Annual Distance: 7,300 km

Result

How much will I save?

CO2 emissions reductions during the lifecycle of the vehicle 4.81 tonnes.

1.31

CO2 emissions from conventional vehicle

0.99

CO2 emissions due to electric vehicle

0.32

Annual CO2 emissions savings by opting EV

4.81

CO2 emissions reductions during the lifecycle of the vehicle

Start a new Calculation

Results Obtained

- CO₂ emissions from conventional vehicle: **1.31 tonnes**
- CO₂ emissions from electric vehicle: **0.99 tonnes**
- Annual CO₂ savings by switching to EV: **0.32 tonnes**
- Total CO₂ reduction over vehicle lifetime: **4.81 tonnes**

AI for Sustainability

What is AI for Sustainability?

Artificial Intelligence (AI) helps solve environmental problems by analyzing data, automating tasks, and making systems more efficient.

Applications of AI

Smart Waste Management

- AI-powered smart bins automatically identify and sort waste for recycling.
- Helps reduce landfill waste and improve recycling efficiency.

Smart Agriculture

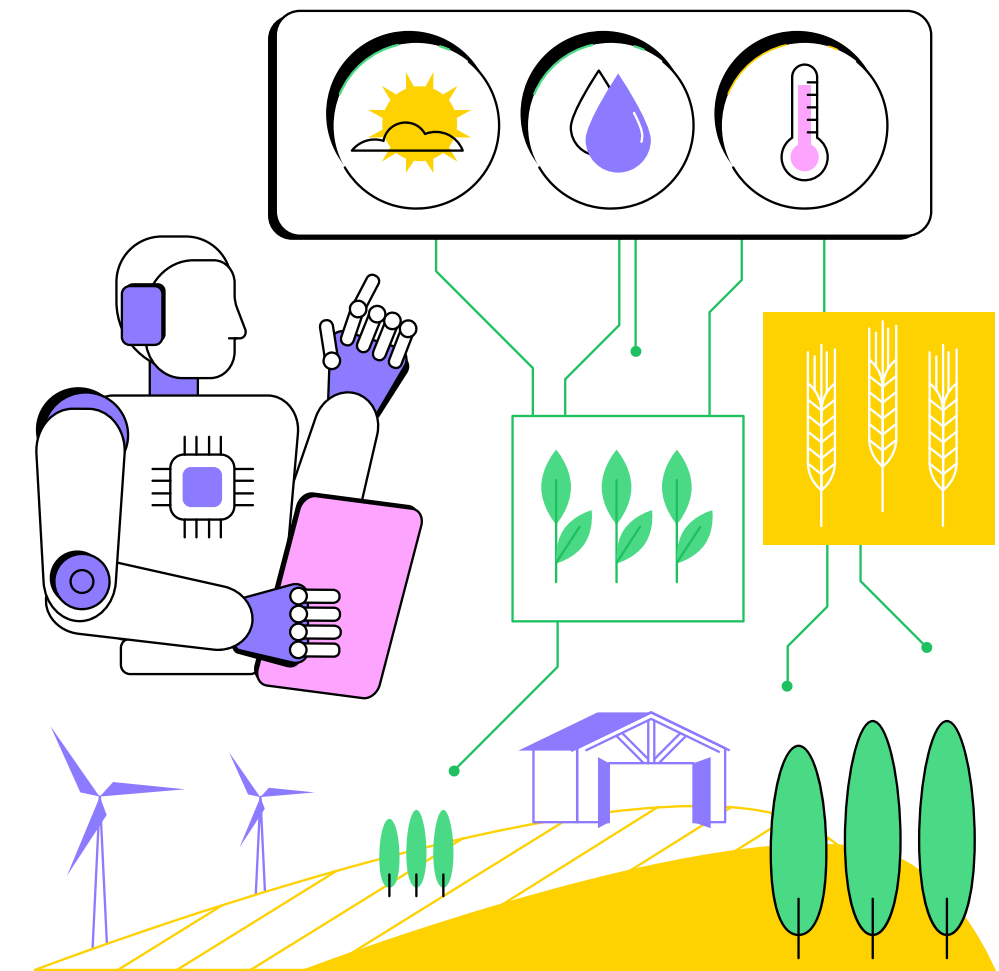
- AI monitors crop health, detects plant diseases, and optimizes irrigation.

Energy Management

- AI predicts electricity demand and optimizes energy consumption.

Smart Transportation

- AI optimizes traffic flow and promotes electric vehicle



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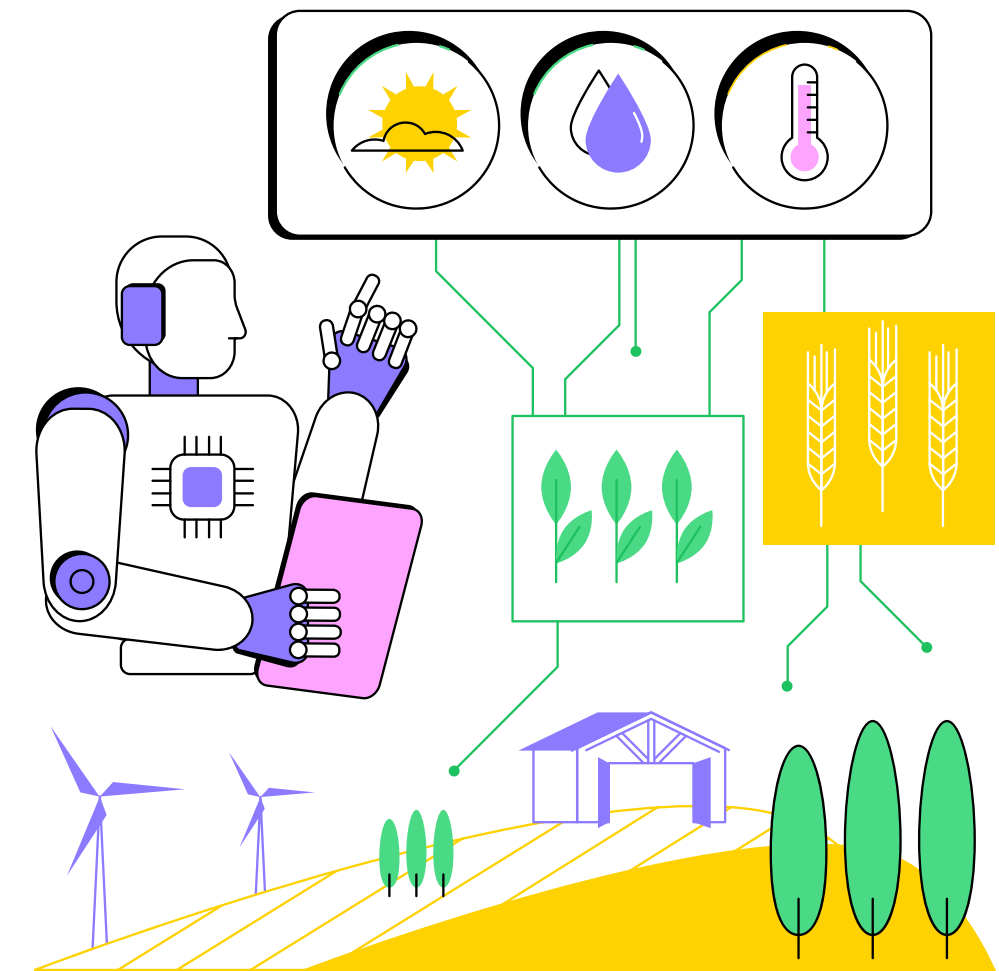
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Benefits and Challenges of AI for Sustainability



Reduces energy consumption through smart optimization.



Improves waste management and recycling.



Supports precision agriculture and saves water.



Helps monitor air quality, pollution, and climate change

Project

EcoSort AI – Smart Waste Classification

Project Objective

Develop an AI-powered system that automatically identifies and classifies waste into categories such as Plastic, Paper, Glass, Metal, Organic Waste, and E-waste using image recognition.

Problem Statement

Improper waste segregation leads to pollution, inefficient recycling, and increased landfill waste.

Proposed Solution

Use an AI model to analyze an image of waste and automatically classify it into the correct waste category. The system will also suggest the appropriate recycling or disposal method.



Thank You